Quadratic Equations - VDPP

Questions

1. Find the remainder when $p(x) = x^3 + 4x^2 - 2x + 5$ is divided by $g(x) = x - 1$

(a) 8

(b) −8

(c) 2

(d) −2

2. If $p(x) = 3x^3 - 5x^2 - 11x - 3$ is a polynomial of degree three then find the relationship between zeros and the coefficients.

(a) $\alpha + \beta + \gamma = \frac{5}{3}$

(b) $\alpha\beta + \beta\gamma + \gamma\alpha = -\frac{11}{3}$

(c) $\alpha\beta\gamma = 1$

(d) All of the above

3. Find the value of a, if x − a is a factor of $x^3 - a^2x + x + 2$.

(a) −2

(b) −3

(c) 2

(d) 3

4. Determine the value of k, which the polynomial $2x^4 - kx^2 + 2x + 1$ is divisible by 1 − 2x.

(a) $-\frac{17}{2}$

(b) $\frac{17}{2}$

(c) 17

(d) -17

5. Find the value of a, if x − a is factor of $x^3 - a^2x + x + 2$.

(a) −2

(b) −3

(c) 2

(d) 3

6. Determine the value of k, for which the polynomial $2x^4 - kx^3 + 4x^2 + 2x + 1$ is divisible by 1 – 2x.

(a) –25

(b) –30

(c) 25

(d) 30

7. What will be the remainder when polynomial $f(x) = x^3 + x^2 - 3x + 2$ is divided by x + 1.

(a) 0

(b) –5

(c) –1

(d) 5

8. If $f(x) = x^2 - 3x + 4$, then find the values of x satisfying the equation f(x) = f(2x + 1)

(a) $1, \frac{2}{3}$

(b) $-1, \frac{2}{3}$

(c) $-1, \frac{-2}{3}$

(d) $1, \frac{-2}{3}$

9. Solve the following quadratic equation $8x^2 + 2x + 1 = 0$

(a) $\frac{-2\pm\sqrt{-36}}{16}$

(b) $\frac{-2\pm\sqrt{-28}}{16}$

(c) $\frac{-2\pm\sqrt{-36}}{8}$

(d) $\frac{-2\pm\sqrt{-28}}{8}$

10. If $f(x) = (x - a)^2(x - b)^2$, find f (a + b).

(a) $a^2 . b^2$

(b) $a^2 + b^2$

(c) $-a^2b^2$

(d) $a^2 - b^2$

11. Solve the following quadratic equation $x^2 + 3ix + 10$

(a) –3i, 7i

(b) –5i, 2i

(c) –3i, 5i

(d) –5i, 3i

12. The coefficient of x in the equation $x^2 + px + q = 0$ was wrongly as 17 in place of 13 and the roots thus found were –2 and –15. Find the roots of correct equation.

(a) 3, –10

(b) –3, –10

(c) –3, 10

(d) None of these

13. If α and β are roots of $ax^2 + 2bx + c = 0$, then $\sqrt{\frac{\alpha}{\beta}} + \sqrt{\frac{\beta}{\alpha}}$ is equal to

(a) $\frac{2b}{\sqrt{ac}}$

(b) $\frac{-2b}{\sqrt{bc}}$

(c) $\frac{-2b}{\sqrt{ac}}$

(d) $\frac{2a}{\sqrt{bc}}$

14. If α and β are the roots of the equation $x^2 - \left(1 + n^2\right)x + \frac{1}{2}\left(1 + n^2 + n^4\right) = 0$, then the value of $\alpha^2 + \beta^2$ is

(a) n

(b) $- n^2$

(c) – n

(d) n^2

15. If the product of roots of the equation $mx^2 + 6x + (2m - 1) = 0$ is –1, then the value of m will be

(a) $\frac{2}{3}$

(b) $\frac{-1}{3}$

(c) $\frac{1}{3}$

(d) $\frac{-2}{3}$

16. If the roots of the equation $ax^2 + bx + c = 0$ are α, β, then the value of $\alpha\beta^2 + \alpha^2\beta + \alpha\beta$ will be

(a) $\frac{a(c-d)}{b^2}$

(b) $\frac{a(b-c)}{c^2}$

(c) $\frac{b(c-a)}{a^2}$

(d) $\frac{c(a-b)}{a^2}$

17. If the roots of the equation $x^2 - 5x + 16 = 0$ are α, β and the roots of equation $x^2 + px + q = 0$ are $\alpha^2 + \beta^2$, $\frac{\alpha\beta}{2}$, then find the value of p and q.

(a) p = –1, q = –56

(b) p = 1, q = 56

(c) p = –56, q = 1

(d) p = –2, q = –56

18. If α and β are the roots of equation $ax^2 + bx + c = 0$ then the value of $(\alpha - \beta)^2$ = ?

(a) $\frac{b^2+ac}{a^2}$

(b) $\frac{b^2-ac}{a^2}$

(c) $\frac{b^2+4ac}{a^2}$

(d) $\frac{b^2-4ac}{a^2}$

19. Find the condition that the roots of the equation $ax^2 + bx + c = 0$ should be: equal in magnitude and opposite in sign;

(a) a < 0 , c < 0

(b) a > 0 , b = 0, c < 0

(c) a > 0 , b = 0, c > 0

(d) a > 0 , b > 0, c < 0

20. Find the condition that the roots of the equation $ax^2 + bx + c = 0$ should be reciprocal to each other

(a) a = c

(b) a = b

(c) b = c

(d) a = -b

21. The root of the quadratic equation $(a + b - 2c) x^2 - (2a - b - c) x + (a - 2b + c) = 0$ are:

(a) $a + b + c$ & $a - b + c$

(b) $1/2$ & $a - 2b + c$

(c) $a - 2b + c$ & $1/(a + b - 2c)$

(d) None of these

22. If A.M. of the roots of a quadratic equation is 8/5 and A.M. of their reciprocals is 8/7, then equation is?

(a) $3x^2 - 12x + 7 = 0$

(b) $7x^2 - 16x + 8 = 0$

(c) $5x^2 - 16x + 7 = 0$

(d) $7x^2 - 16x + 5 = 0$

23. If $\sin \alpha$ & $\cos \alpha$ are the roots of the equation $ax^2 + bx + c = 0$, then

(a) $a^2 - b^2 + 2ac = 0$

(b) $a^2 + b^2 + 2ac = 0$

(c) $a^2 - b^2 - 2ac = 0$

(d) $a^2 + b^2 - 2ac = 0$

24. Let $p(x) = 2x^2 + 3bx + c$ and $p(1) = 0$ and $p(2) = 0$ the find $p(3)$.

(a) 3

(b) 4

(c) −4

(d) 5

25. If α, β are the roots of the equation $ax^2 + bx + c = 0$, then the equation whose roots are $\alpha + \frac{1}{\beta}\ and\ \beta + \frac{1}{\alpha}$ is

(a) $acx^2 - (a + c) x + (a + c)^2 = 0$

(b) $acx^2 + (a + c) x - (a + c)^2 = 0$

(c) $acx^2 + (a + c) bx + (a + c)^2 = 0$

(d) $acx^2 - (a + c)bx + (a + c)^2 = 0$

26. Let $\alpha \neq \beta$ satisfy $\alpha^2 + 1 = 6\alpha$, $\beta^2 + 1 = 6\beta$. Then, the quadratic equation whose roots are $\frac{\alpha}{\alpha+1}, \frac{\beta}{\beta+1}$ is

(a) $4x^2 - 4x + 1 = 0$

(b) $8x^2 + 8x - 1 = 0$

(c) $8x^2 - 8x - 1 = 0$

(d) $8x^2 - 8x + 1 = 0$

27. If α, β are the roots of the equation $2x^2 - 3x - 6 = 0$, find the equation whose roots are $\alpha^2 + 2$ and $\beta^2 + 2$.

(a) $4x^2 - 49x + 118 = 0$

(b) $4x^2 + 49x + 118 = 0$

(c) $4x^2 - 49x - 118 = 0$

(d) $4x^2 + 49x - 118 = 0$

28. A quadratic equation with rational coefficients one of whose roots in $\tan\left|\frac{\pi}{12}\right|$ is

(a) $x^2 + 2x + 1 = 0$

(b) $x^2 - 4x + 1 = 0$

(c) $x^2 - 2x + 4 = 0$

(d) $x^2 - 4x - 1 = 0$

29. If $\alpha \neq \beta$ but $\alpha^2 = 5\alpha - 3$, $\beta^2 = 5\beta - 3$, then the equation whose roots are $\frac{\alpha}{\beta} and \frac{\beta}{\alpha}$ is

(a) $3x^2 - 19x - 3 = 0$

(b) $3x^2 + 19x + 3 = 0$

(c) $3x^2 - 19x + 3 = 0$

(d) None of these

30. Find the quadratic equation with rational coefficients one of whose roots is $\frac{1}{2+\sqrt{5}}$

(a) $x^2 + 4x - 1 = 0$

(b) $x^2 + 4x + 1 = 0$

(c) $x^2 - 4x - 1 = 0$

(d) $\sqrt{2}x^2 - 4x + 1 = 0$

31. Find the quadratic equation with rational coefficients one of whose roots is $2 + \sqrt{3}$ and hence find the value of expression: $x^3 - 7x^2 + 13x - 2$ for $x = 2 + \sqrt{3}$.

(a) $x^2 + 4x - 1 = 0$

(b) $x^2 + 4x + 1 = 0$

(c) $x^2 - 4x + 1 = 0$

(d) $\sqrt{2}x^2 - 4x + 1 = 0$

32. Find the quadratic equation with real coefficients which has –2 + i as a root (i = $\sqrt{-1}$).

(a) $x^2 - 4x + 5$

(b) $x^2 + 4x + 5$

(c) $x^2 - 4x - 5$

(d) $x^2 + 4x - 5$

33. Form an equation whose roots are 2, and $-\frac{1}{2}$.

(a) $2x^2 + 3x - 2 = 0$

(b) $3x^2 - 2x + 2 = 0$

(c) $2x^2 - 3x - 2 = 0$

(d) $3x^2 + 2x + 2 = 0$

34. The quadratic equation whose roots are $\alpha^2 + \alpha\beta$ & $\beta^2 + \alpha\beta$ is:

(a) $x^2 + (\alpha^2 + \beta^2 + 2\alpha\beta)\, x - (\alpha^2 + \alpha\beta)(\beta^2 + \alpha\beta)$

(b) $x^2 - (\alpha^2 + \beta^2 + 2\alpha\beta)\, x - (\alpha^2 + \alpha\beta)(\beta^2 + \alpha\beta)$

(c) $x^2 - (\alpha^2 + \beta^2 + 2\alpha\beta)\, x + (\alpha^2 + \alpha\beta)(\beta^2 + \alpha\beta)$

(d) $x^2 + (\alpha^2 + \beta^2 + 2\alpha\beta)\, x + (\alpha^2 + \alpha\beta)(\beta^2 + \alpha\beta)$

35. The quadratic equation $x^2 + 7x - 14(q^2 + 1) = 0$, where q is an integer has

(a) Real and distinct roots

(b) No real roots

(c) Equal roots

(d) None

36. Which of the following is/are always false?

(a) A quadratic equation with rational co-efficients has zero or two irrational roots.

(b) A quadratic equation with real coefficients has zero or two non-real roots.

(c) A quadratic equation with irrational coefficients has zero or two rational roots.

(d) A quadratic equation with integer coefficients has zero or two irrational roots.

37. If l, m, n are real $l \neq m$, then the roots of the equation

$(l - m)\, x^2 - 5\,(l + m)\, x - 2\,(l - m) = 0$ are

(a) complex

(b) real and equal

(c) real and distinct

(d) none of these

38. The condition for the roots of the equation, $(c^2 - ab)\, x^2 - 2\,(a^2 - bc)\, x + (b^2 - ac) = 0$ to be equal is

(a) $a = 0$

(b) $b = 0$

(c) $c = 0$

(d) None of these

39. If $a + b + c = 0$ then the roots of the equation $4ax^2 + 3bx + 2c = 0$ where $a, b, c \in R$ are

(a) Real and distinct roots

(b) No real roots

(c) Equal roots

(d) None

40. Find the nature of the roots of equation $2x^2 + x + 3 = 0$

(a) real and distinct roots

(b) No real roots

(c) Equal roots

(d) None

41. Number of non-zero real roots of the equation $\sqrt{x}+\sqrt{x-\sqrt{(1-x)}}=1$ is

(a) 0

(b) 1

(c) 2

(d) 3

42. If the roots of the equation $x^2 - 8x + (a^2 - 6a) = 0$ are real, then

(a) $-2 < a < 8$

(b) $2 < a < 8$

(c) $-2 \leq a \leq 8$

(d) $2 \leq a \leq 8$

43. The number of real roots of the equation $x^2 - 3|x| + 2 = 0$ is

(a) 4

(b) 3

(c) 2

(d) 1

44. The number of positive integral solution of the inequation $\frac{x^2(3x-4)^3(x-2)^4}{(x-5)^5(2x-7)^6} \leqslant 0$ is

(a) 2

(b) 0

(c) 3

(d) 4

45. The values of 'a' for which the roots of the equation $(a + 1)x^2 - 3ax + 4a = 0$ $(a \neq -1)$ to be greater than unity are

(a) $0 < a < 1$

(b) $-\frac{16}{7} \leqslant a < -1$

(c) $-2 < a < -1$

(d) None of these

46. If the quadratic equation $2x^2 + ax + b = 0$ and $2x^2 + bx + a = 0$, $(a \neq b)$ have a common root, the value of $a + b$ is

(a) 2

(b) –1

(c) –3

(d) –2

47. If the two equations $x^2 - cx + d = 0$ and $x^2 - ax + b = 0$ have common root and the second has equal roots, then 2(b + d) =

(a) ac

(b) ab

(c) bc

(d) bd

48. If $3x^2 + 4x + 6 = 0$ and $9x^2 + 12x + 18 = 0$ are two equations, then they have ____ common roots.

49. If $x^2 + ax + 10 = 0$ and $x^2 + bx - 10 = 0$ have a common root, then $a^2 - b^2$ is equal to

(a) 10

(b) 20

(c) 30

(d) 40

50. $x^2 - 11x + a$ and $ax^2 - 14x + 2a$ will have a common factor, if a =

(a) 24

(b) 0, 24

(c) 3, 24

(d) 0, 3

51. If one root of the equation $x^2 + px + 12 = 0$ is 4, while the equation $x^2 + px + q = 0$ has equal roots, then the value of q is

(a) 49/4

(b) 4/49

(c) 4

(d) none of these

52. If the roots of the equation $x^3 + Px^2 + Qx - 19 = 0$ are each one more than the roots of the equation $x^3 - Ax^2 + Bx - C = 0$, where A, B, C, P & Q are constants then the value of A + B + C =

(a) 18

(b) 19

(c) 20

(d) None

53. The equation $x^3 - ax + b = 0$ & $x^3 - px^2 = 0$, where $b \neq 0$, $q \neq 0$, have one common root & the second equation has two equal roots then k(q + b) = ap, where the value of k is__.

54. Statement-I: If equation $ax^2 + bx + c = 0$; (a, b, c ∈ R) and $2x^2 + 3x + 4 = 0$ have a common root, then a : b : c = 2 : 3 : 4.

Statement-II: If p + iq is one root of a quadratic equation with real coefficients then p – iq will be the other root : p, q ∈ R, $i - \sqrt{-1}$

(a) Statement I is true, Statement II is true; Statement II is correct explanation for Statement I.

(b) Statement I is true, Statement II is true; Statement II is NOT a correct explanation for Statement-I.

(c) Statement I is true, Statement II is false

(d) Statement I is false, Statement II is true.

55. The non-zero value of a for which the equations $x^2 - 11x + a$ and $x^2 - 14x + 2a$ have a common factor is _____.

56. If $x^2 + 3x + 5 = 0$ and $ax^2 + bx + c = 0$ have a common root and a, b, c ∈ N, the minimum value of a + b + c is

(a) 3

(b) 9

(c) 6

(d) 12

57. The range of $-2x^2 + x + 3$ is:

(a) $\left(\frac{25}{8}, \infty\right)$

(b) $\left(-\infty, \frac{25}{8}\right]$

(c) $\left[\frac{25}{8}, \infty\right)$

(d) $\mathbb{R} - \left\{\frac{25}{8}\right\}$

58. $x^2 + 2kx + 10 \geq 0$ $\forall x$ then

(a) $k \in [-10, 10]$

(b) $k \in \left[-\sqrt{10}, \sqrt{10}\right]$

(c) $k \in [-4, 4]$

(d) None

59. Find the value of 'a' for which $x^2 + 2ax + 8 + 2a > 0$, for all $x \in \mathbb{R}$.

(a) $-3 < a < 4$

(b) $3 < a < 5$

(c) $-2 < a < 4$

(d) $2 < a < 4$

60. For all $x \in \mathbb{R}$. If $\lambda x^2 - 7\lambda x + 8\lambda + 1 > 0$ then, Find the values of λ.

(a) $0 \leqslant \lambda < \frac{17}{4}$

(b) $0 \leqslant \lambda < \frac{4}{17}$

(c) $0 \leqslant \lambda < \frac{4}{15}$

(d) $0 < \lambda < \frac{3}{17}$

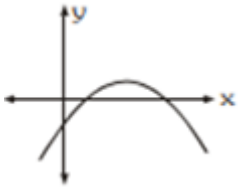
61. Graph of the function $f(x) - Ax^2 - Bx + C$, where

$A = (\sec\theta - \cos\theta)(\operatorname{cosec}\theta - \sin\theta)(\tan\theta + \cot\theta)$,

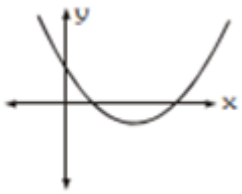
$B = (\sin\theta + \operatorname{cosec}\theta)^2 + (\cos\theta + \sec\theta)^2 - (\tan^2\theta + \cot^2\theta)$ &

$C = 12$, is represented by

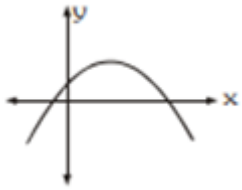
(a)



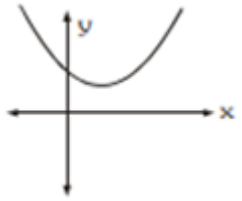
(b)



(c)

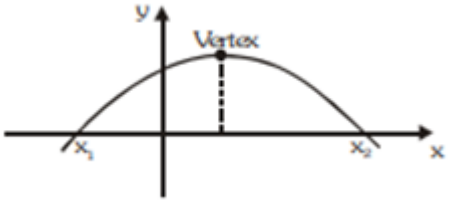


(d)



62.

The adjoining figure shows the graph of $y = ax^2 + bx + c$. Then,



(a) $a > 0$

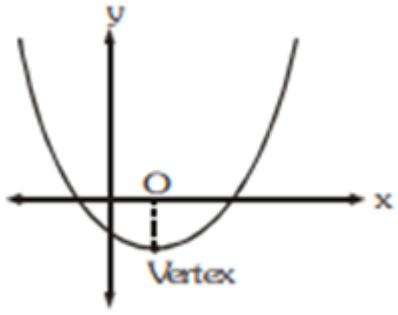
(b) $b > 0$

(c) $c > 0$

(d) $b^2 < 4ac$

63.

Graph of $y = ax^2 + bx + c = 0$ is given adjacently. What conclusions can be drawn from this graph:



(a) $a > 0$

(b) $b^2 - 4ac > 0$

(c) $c < 0$

(d) All of the above

64. If the roots of the equation: $3x^2 + 2(k^2 + 1)x + (k^2 - 3k + 2) = 0$ be of opposite signs, then

(a) $k = 1$

(b) $1 < k < 2$

(c) $k = 2$

(d) $-2 < k < -1$

65. If 2 lies between the roots of quadratic equation $x^2 - ax + a = 0$, then:

(a) $0 < a < 4$

(b) $a < 4$

(c) $a > 4$

(d) None of these

66. If the roots of the equation $x^2 - 2ax + a^2 + a - 3 = 0$ are real and greater than 3, then:

(a) $a \in \varphi$

(b) $a > 3$

(c) $a < 2$ or $a > 3$

(d) $a \leq 3$

67. Let S be the set of values of 'a' for which 2 lies between the roots of quadratic equation $x^2 + (a + 2)x - a - 3 = 0$. Then S is given by

(a) $(-\infty, -5)$

(b) $(5, \infty)$

(c) $(-\infty, 5)$

(d) $(-5, \infty)$

68. The values of a for which both the roots of the equation $x^2 - (2a - 1)x + a = 0$ are positive is:

(a) $\left(\frac{2-\sqrt{3}}{2}, \infty\right)$

(b) $\left(\frac{2-\sqrt{3}}{2}, \frac{2+\sqrt{3}}{2}\right)$

(c) $\left(\frac{2+\sqrt{3}}{2}, \infty\right)$

(d) None of these

69. The values of a for which one root is greater than 1 and the other root less than zero of the equation $x^2 + (2a + 1)x + (a - 1) = 0$ is:

(a) $a < -\frac{1}{3}$

(b) $a > 0$

(c) $a < a < 1$

(d) $a < 1$

70. If the roots of the equation $x^2 - 2ax + a^2 + a - 3 = 0$ are real less than 3, then:

(a) $a < 2$

(b) $2 \leq a \leq 3$

(c) $3 < a \leq 4$

(d) $a > 4$

71. Find the values of k such that $x^2 + 2x + k = 0$ has roots such that 3 lies outside the roots.

(a) $K \in (-15, 3]$

(b) $K \in (-15, 5]$

(c) $K \in (-15, 1]$

(d) $K \in (-15, -1]$

72. The values of m, for which both the roots of equation $x^2 - 2mx + m^2 - 1$ are greater than -2 but less than 4 is-

(a) $(-1, 3)$

(b) $(-2, 4)$

(c) $(-\infty, 3) \cup (5, \infty)$

(d) None of these

73. If α, β are the roots of the quadratic equation $(p^2 + p + 1)x^2 + (p - 1)x + p^2 = 0$ such that unity lies between the roots then the set of values of p is:

(a) φ

(b) $p \in (-\infty, -1) \cup (0, \infty)$

(c) $p \in (-1, 0)$

(d) $(-1, 1)$

74. If α and β and β ($\alpha < \beta$), are the roots of the equation $x^2 + bx + c = 0$, where $c < 0 < b$, then

(a) $0 < \alpha < \beta$

(b) $\alpha < 0 < \beta < |\alpha|$

(c) $\alpha < \beta < 0$

(d) $\alpha < 0 < |\alpha| < \beta$

75. The value of p for which both the roots of the equation $4x^2 - 20px + (25p^2 + 15p - 66) = 0$, are less than 2, lies in

(a) $\left(\frac{4}{5}, 2\right)$

(b) $(2, \infty)$

(c) $\left(-1, -\frac{4}{5}\right)$

(d) $(-\infty, -1)$

76. A car travels 25 km an hour faster than a bus for a journey of 500 km. The bus takes 10 h more than the car. If speed of car is p km/h and speed of bus is q km/h, then

(a) $p = q^2$

(b) $p = 2q$

(c) $p = 3q$

(d) $p^2 = q$

77. If a, b and c are in A.P. and one root of the equation $ax^2 + bx + c = 0$ is 2, then the other root is-

(a) $\frac{2}{3}$

(b) $\frac{-5}{4}$

(c) $\frac{3}{2}$

(d) $\frac{-4}{5}$

78. If α, β are the roots of $ax^2 + bx + c = 0$ and $\alpha + \beta$, $\alpha^2 + \beta^2$, $\alpha^3 + \beta^3$ are in G.P., where $\Delta = b^2 - 4ac$, then

(a) $\Delta \neq 0$

(b) $b\Delta = 0$

(c) $cb \neq 0$

(d) $c\Delta = 0$

79. The range of $\frac{x^2-2x+3}{x^2-2x-8}$, $x \in R - \{-2, 4\}$ is

(a) R

(b) $R - \left[-\frac{2}{9}, 1\right]$

(c) $R - \left(-\frac{2}{9}, 1\right]$

(d) $\left[-\frac{2}{9}, 1\right]$

80. The minimum positive value of $\frac{x^2+2x+4}{x+2}$ is

(a) 1

(b) 2

(c) –2

(d) –6

81. Let $(a+\sqrt{b})^{Q(x)} + (a-\sqrt{b})^{Q(x)-2\lambda} = A$, where $\lambda \in N$, $A \in R$ and $a^2 - b = 1$

$\therefore (a+\sqrt{b})(a-\sqrt{b}) = 1 \Rightarrow (a+\sqrt{b}) = (a-\sqrt{b})^{-1} and (a-\sqrt{b}) = (a+\sqrt{b})^{-1}$

i.e. $(a \pm \sqrt{b}) = (a+\sqrt{b})^{\pm 1}\, or\, (a-\sqrt{b})^{x1}$

By substituting $(a+\sqrt{b})^{Q(x)}$ as t in the equation we get a quadratic in t.

Also, $a + ar + ar^2 \ldots\ldots\ldots \infty = \frac{a}{1-r}$ where $-1 < r < 1$

Solution of $(2+\sqrt{3})^{x^2-2x+1} + (2-\sqrt{3})^{x^2-2x-1} = \frac{4}{2-\sqrt{3}}$ are

(a) $1 \pm \sqrt{3}.1$

(b) $1 \pm \sqrt{2}.1$

(c) $1 \pm \sqrt{3}.2$

(d) $1 \pm \sqrt{2}.2$

82. Let $(a+\sqrt{b})^{Q(x)} + (a-\sqrt{b})^{Q(x)-2\lambda} = A$, where $\lambda \in N$, $A \in R$ and $a^2 - b = 1$

$\therefore (a+\sqrt{b})(a-\sqrt{b}) = 1 \Rightarrow (a+\sqrt{b}) = (a-\sqrt{b})^{-1} and (a-\sqrt{b}) = (a+\sqrt{b})^{-1}$

i.e. $(a \pm \sqrt{b}) = (a+\sqrt{b})^{\pm 1}\, or\, (a-\sqrt{b})^{x1}$

By substituting $(a+\sqrt{b})^{Q(x)}$ as t in the equation we get a quadratic in t.

Also, $a + ar + ar^2 \ldots\ldots\ldots \infty = \frac{a}{1-r}$ where $-1 < r < 1$

The number of real solutions of the equation $(15+4\sqrt{14})^t+(15-4\sqrt{14})^t=30$ are-

where t = x^2 – 2|x|

(a) 0

(b) 2

(c) 4

(d) 6

83. Let $(a+\sqrt{b})^{Q(x)}+(a-\sqrt{b})^{Q(x)-2\lambda}=A$, where λ ∈ N, A ∈ R and a^2 – b = 1

$\therefore (a+\sqrt{b})(a-\sqrt{b})=1 \Rightarrow (a+\sqrt{b})=(a-\sqrt{b})^{-1} and (a-\sqrt{b})=(a+\sqrt{b})^{-1}$

i.e. $(a\pm\sqrt{b})=(a+\sqrt{b})^{\pm1}\, or\, (a-\sqrt{b})^{x1}$

By substituting $(a+\sqrt{b})^{Q(x)}$ as t in the equation we get a quadratic in t.

Also a + ar + ar^2 ..... $\infty=\frac{a}{1-r}$ where –1 < r < 1

If $\left(\sqrt{(49+20\sqrt{6})}\right)^{\sqrt{a\sqrt{a\sqrt{a.....\infty}}}}+(5-2\sqrt{6})^{x^2+x-3-\sqrt{x\sqrt{x\sqrt{x.....\infty}}}}=10$ where a = x^2 – 3,

then x is:

(a) $-\sqrt{2}$

(b) $\sqrt{2}$

(c) –2

(d) 2

84. If p & q are distinct reals, then 2{(x – p) (x – q) + (p – q) + (q – x) (q –p)} = $(p-q)^2$ + $(x-p)^2$ + $(x-q)^2$ is satisfied by

(a) no value of x

(b) exactly one value of x

(c) exactly two values of x

(d) infinite values of x

85. The value of 'a' for which the expression $y=x^2+2a\sqrt{a^2-3x+4}$ is square, is:

(a) 4

(b) $\pm\sqrt{3}$

(c) ±2

(d) $a\in\left(-\infty,-\sqrt{3}\right]\cup\left[\sqrt{3},\infty\right)$

86. Find the product of the real roots of the equation, $x^2 + 18x + 30 = 2\sqrt{x^2 + 18x + 45}$.

87. Find the set of values of 'a' for which the quadratic polynomial

$(a + 4)x^2 - 2ax + 2a < 6 < 0 \forall, x \in R$

(a) For all $a \in (-\infty, +\infty)$

(b) For all $a \in (-\infty, 6)$

(c) For all $a \in (-6, -\infty)$

(d) For all $a \in (-\infty, -6)$

88. If α, β, γ be the roots of the equation $ax^3 + bx^2 + cx + d = 0$. To obtain the equation whose roots are $f(\alpha), f(\beta), f(\gamma)$, where f is a function, we put $y = f(\alpha)$ and simplify it to obtain $\alpha = g(y)$ (some function of y). Now, α is a root of the equation $ax^3 + bx^2 + ca + d = 0$, then we obtain the desired equation which is $a\{g(y)\}^3 + b\{g(y)\}^2 + c\{g(y)\} + d = 0$.

For example, If α, β, γ are the roots of $ax^3 + bx^2 + cx + d = 0$. To find equation whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$ we put $y = \frac{1}{\alpha} \Rightarrow \alpha = \frac{1}{y}$

As α is a root of $ax^3 + bx^2 + cx + d = 0$

We get $\frac{a}{y^2} + \frac{b}{y^2} + \frac{c}{y} + d = 0$

$\Rightarrow dy^3 + cy^2 + by + a = 0$

This is desired equation

If α, β are the roots of the equation $ax^2 + bx + c = 0$, then the roots of the equation $a(2x + 1)^2 + b(2x + 1)(x - 1) + c(x - 1)^2 = 0$ are:

(a) $\frac{2\alpha+1}{\alpha-1}, \frac{2\beta+1}{\beta-1}$

(b) $\frac{2\alpha-1}{\alpha+1}, \frac{2\beta-1}{\beta+1}$

(c) $\frac{\alpha+1}{\alpha-2}, \frac{\beta+1}{\beta-2}$

(d) $\frac{2\alpha+3}{\alpha-1}, \frac{2\beta+3}{\beta-1}$

89. If α, β, γ be the roots of the equation $ax^3 + bx^2 + cx + d = 0$. To obtain the equation whose roots are $f(\alpha), f(\beta), f(\gamma)$, where f is a function, we put $y = f(\alpha)$ and simplify it to obtain $\alpha = g(y)$ (some function of y). Now, α is a root of the equation $ax^3 + bx^2 + ca + d$

= 0, then we obtain the desired equation which is a{g(y)}^{3} + b{g(y)}^{2} + c{g(y)} + d = 0.

For example, If α, β, γ are the roots of ax^3 + bx^2 + cx + d = 0. To find equation whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$ we put $y = \frac{1}{\alpha} \Rightarrow \alpha = \frac{1}{y}$

As α is a root of ax^3 + bx^2 + cx + d = 0

We get $\frac{a}{y^2} + \frac{b}{y^2} + \frac{c}{y} + d = 0$

$\Rightarrow$ dy^3 + cy^2 + by + a = 0

This is desired equation

If α, β are the roots of the equation 2x^2 + 4x – 5 = 0, the equation whose roots are the reciprocals of 2α – 3 and 2β – 3 is:

(a) x^2 + 10x – 11 = 0

(b) 11x^2 + 10x + 1 = 0

(c) x^2 + 10x + 11 = 0

(d) 11x^2 – 10x + 1 = 0

90. If α, β, γ be the roots of the equation ax^3 + bx^2 + cx + d = 0. To obtain the equation whose roots are f(α), f(β), f(γ), where f is a function, we put y = f(α) and simplify it to obtain α = g(y) (some function of y). Now, α is a root of the equation ax^3 + bx^2 + ca + d = 0, then we obtain the desired equation which is a{g(y)}^{3} + b{g(y)}^{2} + c{g(y)} + d = 0.

For example, If α, β, γ are the roots of ax^3 + bx^2 + cx + d = 0. To find equation whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$ we put $y = \frac{1}{\alpha} \Rightarrow \alpha = \frac{1}{y}$

As α is a root of ax^3 + bx^2 + cx + d = 0

We get $\frac{a}{y^2} + \frac{b}{y^2} + \frac{c}{y} + d = 0$

$\Rightarrow$ dy^3 + cy^2 + by + a = 0

This is desired equation

If α, β are the roots of the equation px^2 – qx + r = 0, then the equation whose roots are $\alpha^2 + \frac{r}{p}\ and\ \beta^2 + \frac{r}{p}$ is

(a) p^{3}x^2 + pq^{2}x + r = 0

(b) px^2 – qx + r = 0

(c) $p^3x^2 - pq^2x + q^2r = 0$

(d) $px^2 + qx - r = 0$

91. If α, β, γ be the roots of the equation $ax^3 + bx^2 + cx + d = 0$. To obtain the equation whose roots are $f(\alpha), f(\beta), f(\gamma)$, where f is a function, we put $y = f(\alpha)$ and simplify it to obtain $\alpha = g(y)$ (some function of y).

Now, α is a root of the equation $ax^3 + bx^2 + ca + d = 0$, then we obtain the desired equation which is $a\{g(y)\}^3 + b\{g(y)\}^2 + c\{g(y)\} + d = 0$.

For example, If α, β, γ are the roots of $ax^3 + bx^2 + cx + d = 0$. To find equation whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$ we put $y = \frac{1}{\alpha} \Rightarrow \alpha = \frac{1}{y}$

As α is a root of $ax^3 + bx^2 + cx + d = 0$

We get $\frac{a}{y^2} + \frac{b}{y^2} + \frac{c}{y} + d = 0$

$\Rightarrow dy^3 + cy^2 + by + a = 0$

This is desired equation

If α, β, γ are the roots of the equation $x^3 - x - 1 = 0$, then the value of $\prod \left(\frac{1+\alpha}{1-\alpha}\right)$ is equal to:

(a) –7

(b) –5

(c) –3

(d) –1

92. If the quadratic equation $ax^2 + bx + 6 = 0$ does not have two distinct real roots, then the least value of 2a + b is:

(a) 2

(b) –3

(c) –6

(d) 1

Answer Key

1. A	**2.** D	**3.** A	**4.** B	**5.** A	**6.** C
7. D	**8.** B	**9.** B	**10.** A	**11.** B	**12.** B
13. C	**14.** D	**15.** C	**16.** D	**17.** A	**18.** D
19. B	**20.** B	**21.** D	**22.** C	**23.** A	**24.** B
25. C	**26.** D	**27.** A	**28.** B	**29.** C	**30.** A
31. C	**32.** A	**33.** C	**34.** C	**35.** A	**36.** C
37. C	**38.** A	**39.** A	**40.** B	**41.** B	**42.** C
43. A	**44.** C	**45.** B	**46.** D	**47.** A	**48.** 2
49. D	**50.** B	**51.** A	**52.** A	**53.** 2	**54.** A
55. 24	**56.** B	**57.** B	**58.** B	**59.** C	**60.** B
61. B	**62.** C	**63.** D	**64.** B	**65.** C	**66.** A
67. A	**68.** C	**69.** A	**70.** A	**71.** C	**72.** A
73. C	**74.** B	**75.** D	**76.** B	**77.** B	**78.** D
79. B	**80.** B	**81.** B	**82.** C	**83.** D	**84.** D
85. C	**86.** 20	**87.** D	**88.** C	**89.** B	**90.** C
91. D	**92.** B				

Solutions

1. (A)
Method 1: Using division method

$$x-1)x^3+4x^2-2x+5(x^2+5x+3$$
$$x^3-x^2$$
$$-\quad +$$
$$5x^2-2x$$
$$5x^2-5x$$
$$-\quad +$$
$$3x+5$$
$$3x-3$$
$$-\quad +$$
$$8$$

so, $p(x)=\left(x^2+5x+3\right)(g(x))+8$
Method 2: using remainder theorem
$p(1)=(1)^3+4(1)^2-2(1)+5$
$p(1)=1+4-2+5$
$p(1)=8$

2. (D) Given $3x^3-5x^2-11x-3$ is polynomial of degree three and it is of the from

$ax^3+bx^2+cx+d=0.$

$\alpha+\beta+\gamma=\frac{-b}{a}=-\frac{(-5)}{3}$

$\alpha\beta+\beta\gamma+\gamma\alpha=\frac{c}{a}=-\frac{11}{3}$

$\alpha\beta\gamma=-\frac{d}{a}=\frac{3}{3}=1$

3. (A) Let p(x) = $x^3 - a^2x + x + 2$

By factor theorem, (x – a) is a factor of p(x) if p(a) is = 0

∴ p(a) = $a^3 - a^2 \times a + a + 2 = a^3 - a^3 + a + 2 = a + 2$

∴ a + 2 = 0 ⇒ a = –2

4. (B) If (1 – 2x) is a factor of the polynomial $2x^4 + kx^3 + 4x^2 + 2x + 1$,

Then, (1 – 2x) must be the zero of the polynomial.

Putting $x = \frac{1}{2}$ in the polynomial we get,

k = $\frac{17}{2}$

5. (A) Let p(x) = $x^3 - a^2x + x + 2$

By factor theorem, (x – a) is a factor of p(x) if p(a) = o

∴ p(a) = $a^3 - a^2 \times a + a + 2 = a^3 - a^3 + a + 2 = a + 2$

∴ a + 2 = 0 ⇒ a = –2

6. (C) If (1 – 2x) is a factor of the polynomial $2x^4 + kx^3 + 4x^2 + 2x + 1$,

Then, (1 – 2x) must be the zero of the polynomial.

Putting $x = \frac{1}{2}$ in the polynomial we get,

k = 25

7. (D) x = –1, f(x) = $(-1)^3 + (-1)^2 - 3(-1) + 2$

–1 + 1 + 3 + 2 = 5

8. (B) $f(2x + 1) = (2x + 1)^2 - 3(2x + 1) + 4 = 4x^2 - 2x + 2$

For,

f(x) = f(2x + 1) ⇒ $x^2 - 3x + 4 = 4x^2 - 2x + 2$

⇒ $3x^2 + x - 2 = 0$

⇒ $3x^2 + 3x - 2x - 2 = 0$

⇒ (3x – 2) (x + 1) = 0

⇒ $x = -1, \frac{2}{3}$

9. (B) $8x^2 + 2x + 1 = 0$

$x = \frac{-2 \pm \sqrt{4-32}}{16}$

$x = \frac{-2 \pm \sqrt{-28}}{16}$

10. (A) Given $f(x) = (x - a)^2(x - b)^2$

Putting x = a + b we have

$f(a + b) = (a + b - a)^2 (a + b - b)^2$

$= b^2 . a^2$

$f(a + b) = a^2 . b^2$

11. (B) Roots for the quadratic equation $ax^2 + bx + c$, the roots are $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

For the given equation $x^2 + 3ix + 10$,

$b^2 - 4ac = (3i)^2 - 4(1)(10)$

$= 9i^2 - 40$

$= -9 - 40 = -49$

So, the roots are

$x = \frac{-3i \pm \sqrt{-49}}{2}$

$x = \frac{-3i \pm 7i}{2}$

x = –5i, 2i

12. (B) The correct option Is B –3, –10

$x^2 + px + q = 0$

p = 13

q = (–2) × (–15) = 30

$x^2 + 13x + 30 = 0$

(x + 3)(x + 10) = 0

x = –10, –3

13. (C) Since α , β are the roots of $ax^2 + 2bx + c = 0$

$\Rightarrow \alpha + \beta = \frac{-2b}{a} and \, \alpha\beta = \frac{c}{a}$

$\therefore \sqrt{\frac{\alpha}{\beta}} + \sqrt{\frac{\beta}{\alpha}} = \frac{\alpha+\beta}{\sqrt{\alpha\beta}}$

$= \frac{-\frac{2b}{a}}{\sqrt{\frac{c}{a}}} = -\frac{2b}{\sqrt{ac}}$

14. (D) Here $\alpha + \beta = 1 + n^2, \alpha\beta = \frac{1}{2}\left(1 + n^2 + n^4\right)$

$\therefore \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$= \left(1 + n^2\right)^2 - 2.\frac{1}{2}\left(1 + n^2 + n^4\right)$

$= 1 + n^4 + 2n^2 - 1 - n^2 - n^4$

$= n^2$

15. (C) According to the given condition,

$\frac{2m-1}{1} = 1$ ..... $\left[\therefore \; product \; of \; roots = \frac{constant \; term}{coeff. \; of \; x^2}\right]$

⇒ 3m = 1

⇒ $m = \frac{1}{3}$

16. (D) $\alpha + \beta = -\frac{b}{a}\ and\ \alpha\beta = \frac{c}{a}$

∴ α β ^{2} + α ^{2}β + α β = α β (β + α) + α β

= α β (1 + α + β)

$= \frac{c}{a}\left\{1 + \left(-\frac{b}{a}\right)\right\}$

$= \frac{c(a-b)}{a^2}$

17. (A) Since, roots of the equation x^2 – 5x + 16 = 0 are α , β

∴ α + β = 5 and α β = 16

Sum of the roots of the equation x^2 + px + q = 0 is

$\alpha^2 + \beta^2 + \frac{\alpha\beta}{2} = -p$

$\Rightarrow (\alpha + \beta)^2 - 2\alpha\beta + \frac{\alpha\beta}{2} = -p$

⇒ 25 – 32 + 8 = –p

⇒ p = –1

And product of roots is $\left(\alpha^2 + \beta^2\right)\left(\frac{\alpha\beta}{2}\right) = q$

$\Rightarrow \left[(\alpha + \beta)^2 - 2\alpha\beta\right]\left[\frac{\alpha\beta}{2}\right] = q$

$\Rightarrow q = [25 - 32]\frac{16}{2} = -56$

∴ p = –1, q = –56

18. (D) ∴ (α – β)^2 = α ^{2} + β ^{2} – 2α β

= (α + β)^2 – 4α β

$= \left(\frac{-b}{a}\right)^2 - 4\left(\frac{c}{a}\right)$

$= \frac{b^2}{a^2} - \frac{4c}{a}$

$= \frac{b^2 - 4ac}{a^2}$

19. (B) Given equation ax^2 + bx + c = 0

Condition for roots to be equal in magnitude and opposite in sign

Let one root be α then other root is –α .

Sum of roots

$\therefore \alpha + (-\alpha) = \frac{-b}{a}$

$0 = \frac{-b}{a}$

b = 0 ___ (1)

We have one negative root so,

$\therefore \alpha\beta < 0$

$\therefore \frac{c}{a} < 0$

So, here either c or a will be negative.

Means, a and c will be having opposite sign.

a > 0 , c < 0 or c > 0 , a < 0 and b = 0 .

In option (b) we can see option C satisfy all the conditions.

∴ Possible conditions are a > 0 , b = 0, c < 0

20. (B) Let one root be α and other be $\frac{1}{\alpha}$

Thus product of root

$\alpha . \frac{1}{\alpha} = \frac{c}{a}$

$1 = \frac{c}{a}$

a = c

21. (D) Since sum of coefficients = 0

∴ It's one root is 1 and other root is $\frac{a-2b+c}{a+b-2c}$

22. (C) Let α and β are the roots of the equation.

According to the questions,

$\Rightarrow \frac{\alpha+\beta}{2} = \frac{8}{5}$

$\therefore \alpha + \beta = \frac{16}{5}$ .....(1)

According to the equation,

$\Rightarrow \frac{\frac{1}{\alpha}+\frac{1}{\beta}}{2} = \frac{8}{7}$

$\Rightarrow \frac{\alpha+\beta}{2\alpha\beta} = \frac{8}{7}$

⇒ 7(α + β) = 16α β

$\Rightarrow \frac{7\times\frac{16}{5}}{16} = \alpha\beta$

$\therefore \alpha\beta = \frac{7}{5} \ldots\ldots (2)$

Now, new equation,

⇒ x^2 − (α + β)x + (α .β) = 0

From (1) and (2),

$\Rightarrow x^2 - \frac{16}{5}x + \frac{7}{5} = 0$

$\Rightarrow 5x^2 - 16x + 7 = 0$

23. (A) sin α and cos α are the roots of $ax^2 + bx + c = 0$

$\Rightarrow \sin\alpha + \cos\alpha = \frac{-b}{a}$ and $\sin a \cos a = \frac{c}{a}$

$(\sin\alpha + \cos\alpha)^2 = \sin^2\alpha + \cos^2\alpha + 2\sin\alpha\cos\alpha$

$\Rightarrow \frac{b^2}{a^2} = 1 + \frac{2c}{a}$

$\Rightarrow b^2 = a^2 + 2ac$

$\therefore a^2 - b^2 + 2ac = 0$

Hence, option A.

24. (B) $p(x) = 2x^2 + 3bx + c$

Since $p(1) = 0$, $p(2) = 0$

We have,

$2 + 3b + c = 0$ and $8 + 6b + c = 0$

solving both

$-3b = 6$

$b = -2$

Also $2 + 3(-2) + c = 0$

$c = 4$

$\therefore p(x) = 2x^2 - 6x + 4$

$p(3) = 2(3)^2 - 6(3) + 4$

$p(3) = 18 - 18 + 4$

$p(3) = 4$

25. (C) Since, α and β are the roots of $ax^2 + bx + c = 0$

$\therefore \alpha + \beta = -\frac{b}{a}\ and\ \alpha\beta = \frac{c}{a}$

Now, $\left(\alpha + \frac{1}{\beta}\right) + \left(\beta + \frac{1}{\alpha}\right)$

$= (\alpha + \beta) + \frac{\alpha+\beta}{\alpha\beta}$

$= -\frac{b}{ac}(a + c)$

Also, $\left(\alpha+\frac{1}{\beta}\right)\left(\beta+\frac{1}{\alpha}\right)$

$=\alpha\beta+1+1+\frac{1}{\alpha\beta}$

$=2+\frac{c}{a}+\frac{a}{c}$

$=\frac{2ac+c^2+a^2}{ac}=\frac{(a+c)^2}{ac}$

∴ the required equation is

$x^2-\left[\left(\alpha+\frac{1}{\beta}\right)+\left(\beta+\frac{1}{\alpha}\right)\right]x+\left(\alpha+\frac{1}{\beta}\right)\left(\beta+\frac{1}{\alpha}\right)=0$

$x^2+\frac{b}{ac}(a+c)x+\frac{(a+c)^2}{ac}=0$

⇒ $acx^2 + (a + c)bx + (a + c)^2 = 0$

26. (D) Since α , β are the roots of $x^2 - 6x + 1 - 0$.

∴ α + β = 6 and α β = 1

$x^2-\left(\frac{\alpha}{\alpha+1}+\frac{\beta}{\beta+1}\right)x+\left(\frac{\alpha}{\alpha+1}\right)\left(\frac{\beta}{\beta+1}\right)=0$

$\Rightarrow x^2-\left[\frac{(\alpha+\beta)+2\alpha\beta}{(\alpha+\beta)+\alpha\beta+1}\right]x+\frac{\alpha\beta}{(\alpha+\beta)+\alpha\beta+1}=0$

$\Rightarrow x^2-\left[\frac{6+2(1)}{6+1+1}\right]x+\frac{1}{6+1+1}=0$

$\Rightarrow x^2-x+\frac{1}{8}=0$

⇒ $8x^2 - 8x + 1 = 0$

27. (A) Since α , β are the roots of the equation $2x^2 - 3x - 6 = 0$, so

$\alpha+\beta=\frac{3}{2}\ and\ \alpha\beta=-3$

⇒ $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = \frac{9}{4}+6=\frac{33}{4}$

Now,

$(\alpha^2 + 2) + (\beta^2 + 2) = (\alpha^2 + \beta^2) + 4$

$=\frac{33}{4}+4=\frac{49}{4}$

And

$(\alpha^2 + 2)(\beta^2 + 2) = \alpha^2\beta^2 + 2(\alpha^2 + \beta^2) + 4$

$=(3)^2+2\left(\frac{33}{4}\right)+4=\frac{59}{2}$

So, the equation whose roots are $\alpha^2 + 2$ and $\beta^2 + 2$ is

$x^2 - x[(\alpha^2 + 2) + (\beta^2 + 2)] + (\alpha^2 + 2)(\beta^2 + 2) = 0$

$\Rightarrow x^2 - \frac{49}{4}x + \frac{59}{2} = 0$

$\Rightarrow 4x^2 - 49x + 118 = 0$

28. (B) Given one root is $\tan\left(\frac{\pi}{12}\right)$.

We have $\tan\left(\frac{\pi}{12}\right) = 2 - \sqrt{3}$.

So, the other root is $2 + \sqrt{3}$.

Now, sum of roots $= 2 + \sqrt{3} + 2 - \sqrt{3} = 4$ and product of roots = 1

So, the required equation is $x^2 - 4x + 1 = 0$

29. (C) Given, $\alpha \neq \beta$, but $\alpha^2 = 5\alpha - 3$, $\beta^2 = 5\beta - 3$,

By virtue of the given relations we can say that α and β satisfy $x^2 = 5x3$ or they are the

roots of $x^2 - 5x + 3 = 0$

$\therefore \alpha + \beta = 5$ and $\alpha\beta = 3$

So, equation whose roots are $\frac{\alpha}{\beta}\ and\ \frac{\beta}{\alpha}$.

Sum of roots are, $S = \frac{\alpha^2+\beta^2}{\alpha\beta}$

$S = \frac{(\alpha+\beta)^2-2\alpha\beta}{\alpha\beta}$

$S = \frac{5^2-2\times3}{3}$

$S = \frac{19}{3}$

and product is 1

Equation is $x^2 - Sx + P = 0$

$\Rightarrow x^2 - \frac{19}{3}x + 1 = 0$

30. (A) Given root

$\frac{1}{2+\sqrt{5}} = \frac{2-\sqrt{5}}{-1} = -2 + \sqrt{5}$

Therefore, other root is $-2 - \sqrt{5}$

Again, sum of roots = –4 and product of roots = –1.

Hence, the required equation is $x^2 + 4x - 1 = 0$.

31. (C) For the quadratic equation of one root is $2 + \sqrt{3}$

Other root is $2 - \sqrt{3}$

Sum of roots $= 2 + \sqrt{3} + 2 - \sqrt{3} = 4$

Product of roots $= (2+\sqrt{3})(2-\sqrt{3}) = 4-3 = 1$

∴ Quadratic equation is x^2 – 4x +1 = 0

32. (A) According to the problem, coefficients of the required quadratic equation are real and its one root is –2 + i.

We know in a quadratic with real coefficients imaginary roots occur in conjugate pairs).

Since equation has rational coefficients, the other root is –2 –i.

Now, the sum of the roots of the given equation S = (–2 + i) + (–2 –i) = –4.

Product of the roots, P = (–2 + i) (–2 – i) = $(–2)^2$ – i^2 = 4 – (–1) = 4 + 1 = 5

Hence, the required equation is x^2 – sx + P = 0 i.e., x^2 – 4x + 5 = 0.

33. (C) The given roots are and $-\frac{1}{2}$.

Therefore, sum of the roots, $S = 2 + \left(-\frac{1}{2}\right) = \frac{3}{2}$

And the product of the given roots, $P = 2 - \frac{1}{2} = -1$

Therefore, the required equation is x^2 – sx + p

i.e.,, x^2 – (sum of the roots) + product of the roots = 0

i.e., $x^2 = \frac{3}{2}x - 1 = 0$

i.e., $2x^2$ – 3x – 2 = 0

34. (C) Consider the given roots.

$\alpha^2 + \alpha\beta$ and $\beta^2 + \alpha\beta$

We know that

x^2 – (sum of roots) x + product of roots = 0

$x^2 – (\alpha^2 + \alpha\beta + \beta^2 + \alpha\beta)\, x + (\alpha^2 + \alpha\beta)(\beta^2 + \alpha\beta) = 0$

$x^2 – (\alpha^2 + \beta^2 + 2\alpha\beta)x + (\alpha^2 + \alpha\beta)(\beta^2 + \alpha\beta) = 0$

35. (A) $x^2 + 7x – 14\,(q^2 + 1) = 0$

$D = b^2 – 4ac$

$D = 49 + 56(q^2 + 1) > 0$

∴ The equation has real and distinct roots

36. (C) A quadratic equation with irrational coefficients has zero or two rational roots.

37. (C) Its discriminant D = $25(l + m)^2 + 8(l - m)^2$ D > 0

Which is positive, since l, m, n are real and $l \neq m$.

Hence, roots are real and distinct.

38. (A) Roots are equal, if $b^2 - 4ac = 0$

$4(a^2 - bc)^2 - 4(c^2 - ab)(b^2 - ac) = 0$

$\Rightarrow a(a^3 + b^3 + c^3 - 3abc) = 0$

$\Rightarrow a = 0$ or $a^3 + b^3 + c^3 = 3abc$

39. (A) For given equation $4ax^2 + 3bx + 2c = 0$

We have $D = (3b)^2 - 4(4a)(2c)$

$= 9b^2 - 32ac$

$= 9(-a - c)^2 - 32ac$

$= 9c^2\left(\left(\frac{a}{c}\right)^2 - \frac{14}{9}\frac{a}{c} + 1\right)$

$= p\left(\left(\frac{a}{c} - \frac{7}{9}\right)^2 - \frac{49}{81} + 1\right)$

Which is always positive

Hence roots are real and distinct.

40. (B) Given $2x^2 + x + 3 = 0$

$D = b^2 - 4ac$

$D = 1 - 4(2)(3)$

$D = 1 - 24 = -23 < 0$

$\therefore$ The roots are imaginary

41. (B) We have, $\sqrt{x} + \sqrt{x - \sqrt{(1-x)}} = 1$

$\Rightarrow \sqrt{x - \sqrt{(1-x)}} = 1 - \sqrt{x}$

On squaring both sides, we get

$x - \sqrt{1-x} = 1 + x - 2\sqrt{x}$

$\Rightarrow -\sqrt{1-x} = 1 - 2\sqrt{x}$

Again, squaring on both sides, we get

$1 - x = 1 + 4x - 4\sqrt{x}$

$4\sqrt{x} = 5x$

$\Rightarrow \sqrt{x} = \frac{4}{5}$

$\Rightarrow x = \frac{16}{25}$ [on squaring both sides]

Hence, the number of real solution is 1.

42. (C) Roots of $x^2 - 8x + (a^2 - 6a) = 0$ are real.

So, $D \geq 0$

$\Rightarrow 64 - 4(a^2 - 6a) \geq 0$

$\Rightarrow 16 - a^2 + 6a \geq 0$

$\Rightarrow a^2 - 6a - 16 \leq 0$

$\Rightarrow (a - 8)(a + 2) \leq 0$

$\Rightarrow -2 \leq a \leq 8$

43. (A) Given equation:

$x^2 - 3|x| + 2 = 0$

When $x > 0$, we have

$x^2 - 3x + 2 = 0$

$\Rightarrow (x - 2)(x - 1) = 0$

$\Rightarrow x = 2$ or $x = 1$

and when $x < 0$

$x^2 + 3x + 2 = 0$

or, $(x + 2)(x + 1) = 0$

or, $x = -2$ or $x = -1$

Therefore, number of real roots is 4.

44. (C) Given $\frac{x^2(3x-4)^3(x-2)^4}{(x-5)^5(2x-7)^6} \leqslant 0$

$\Rightarrow \frac{3x-4}{2x-7} \leqslant 0$ and $x = 0, 2, \neq 7/2$

$\Rightarrow \frac{4}{3} \leqslant x < 5$ and $x = 0, 2, \neq 7/2$

Hence, the number of integral solution is {0, 2, 3, 4} = 4

And number of positive integral solutions are 2, 3, 4

Thus the number of integral solution are 3

45. (B) Let f(x) = (a + 1) x^2 – 3ax + 4a and let α , β be the roots of the equation f(x) = 0

The equation will have roots greater than 1 if

(1) D $\geq$ 0

(2) $\alpha + \beta > 2$

(3) (a + 1)f(1) > 0

Now D $\geq$ 0 $\Rightarrow$ $9a^2 - 16a(a+1) \geq 0 \Rightarrow -7a^2 - 16a \geq 0$

$\Rightarrow a(7a+16) \leqslant 0 \Rightarrow -\frac{16}{7} \leqslant a \leqslant -o$ .....(1)

$\alpha + \beta > 2 \Rightarrow \frac{3a}{a+1} > 0 \Rightarrow \frac{3a}{a+1} - 2 > 0 \Rightarrow \frac{3a-2a-2}{a+1} > 0$

$\frac{a-2}{a+1} > 0 \Rightarrow a < -1 \; or \; a > 2$....(ii)

(a + 1)f(1) > 0 $\Rightarrow$ (a + 1) (a + 1 – 3a + 4a) > 0

$\Rightarrow$ (a + 1) (2a + 1) > 0 $\Rightarrow$ a < –1 or

$a > -\frac{1}{2}$ .....(iii)

From (i), (ii) and (iii) we get

$-\frac{16}{7} \leqslant a < -1$

46. (D) Subtract one equation from the other

(a – b) x + (b – a) = 0

(a – b) x = (a – b)

x = 1

$\therefore$ 2 + a + b = 0

a + b = –2

47. (A) Let roots of x^2 – cx + d = 0 be α , β

then roots of x^2 – ax + b = 0 be α , α

$\therefore \alpha + \beta = c$, $\alpha \beta = d$, $\alpha + \alpha = \alpha$, $\alpha^2 = b$

Hence 2 (b + d) = 2 ($\alpha^2 + \alpha \beta$) = 2α ($\alpha + \beta$) = ac.

48. (2)

Given $3x^2 + 4x + 6 = 0$ and $9x^2 + 12x + 18 = 0$

Take the ratio of coefficients

$\frac{3}{9} = \frac{4}{12} = \frac{6}{18} = \frac{1}{3}$

∴ The equation has 2 common roots.

49. (D) Let α be a common root, then

$\alpha^2 + a\alpha + 10 = 0$.....(i)

and $\alpha^2 + b\alpha - 10 = 0$.....(ii)

from (i) – (ii),

$(a - b)\alpha + 20 = 0$

$\Rightarrow \alpha = -\frac{20}{a-b}$

Substituting the value of α in (i), we get

$\left(-\frac{20}{a-b}\right)^2 + a\left(-\frac{20}{a-b}\right) + 10 = 0$

$\Rightarrow 400 - 20a(a - b) + 10(a - b)^2 = 0$

$\Rightarrow 40 - 2a^2 + 2ab + a^2 + b^2 - 2ab = 0$

$\Rightarrow a^2 - b^2 = 40$

50. (B) Expressions are $x^2 - 11x + a$ and $x^2 - 14x + 2a$ will have a common factor, then

$\Rightarrow \frac{x^2}{-22a+14a} = \frac{x}{a-2a} = \frac{1}{-14+11}$

$\Rightarrow \frac{x^2}{-8a} = \frac{x}{-a} = \frac{1}{-3}$

$\Rightarrow x^2 = \frac{8a}{3}\ and\ x = \frac{a}{3}$

$\Rightarrow \left(\frac{a}{3}\right)^2 = \frac{8a}{3}$

$\Rightarrow \frac{a^2}{9} = \frac{8a}{3}$

$\Rightarrow a = 0, 24$

51. (A) Given $x^2 + px + 12 = 0$

Since, x = 4 is one root of the equation, therefore x = 4 will satisfy this equation

∴ $16 + 4p + 12 = 0 \Rightarrow p = -7$

Other quadratic equation becomes $x^2 - 7x + q = 0$

(By putting value of p)

Its roots are equal, so $b^2 = 4ac$

$\Rightarrow 49 = 4q\ or\ q = \frac{49}{4}$

52. (A) Let roots of $x^3 - Ax^2 + Bx - C = 0$ are α, β, γ

$\Rightarrow \alpha + \beta + \gamma = A, \sum \alpha\beta = B, \alpha\beta\gamma = C$

& $(\alpha + 1)(\beta + 1)(\gamma + 1) = 19$

$\Rightarrow (\alpha + \beta + \gamma) + (\alpha\beta + \beta\gamma + \gamma\alpha) + \alpha\beta\gamma + 1 = 19$

$A + B + C = 18$

53. (2) Let $x^2 - ax + b = 0$ has roots α & β

& $x(x^2 - px + q) = 0$ has roots $0, \alpha, \alpha$

but $x = 0$ is not the common root (since $b \neq 0$)

& $x^2 - px + q = 0$ have two equal roots

$\Rightarrow D = 0 \Rightarrow p^2 - 4q = 0$ & $\frac{p}{2}$ is its root

$\frac{p}{2}$ satisfy equation $x^2 - ax + b = 0$

$\Rightarrow \frac{p^2}{4} - \frac{ap}{2} + b = 0 \Rightarrow \frac{p^2}{4} + b = \frac{ap}{2}$

$\Rightarrow 2(q + b) = ap \Rightarrow k = 2$

54. (A) Given, $ax^2 + bx + c = 0$, $2x^2 + 3x + 4 = 0$ have a root in common.

Roots of $2x^2 + 3x + 4 = 0$ are $\frac{-3 \pm i\sqrt{23}}{4}$

$\therefore$ Roots of $2x^2 + 3x + 4 = 0$ are complex.

$\therefore$ Common root is also a complex.

a, b, c$\in$ R $\therefore$ If one root is complex, then another root is conjugate of that complex root.

$\therefore$ Both roots of both equations are same.

$\therefore$ a : b : c = 2 : 3 : 4

Assertion is true.

Reason

It is true that is one root is p + iq for a quadratic equation with real coefficients then its conjugate p – iq will be other root. (p, q $\in$ R)

Reason is true and the correct explanation to given assertion.

55. (24) Let x – a is the common factor of given two polynomials.

then, for x = a,

$a^2 - 11a + k = 0$ .....(1)

$a^2 - 14a + 2k = 0$ .....(2)

Subtracting (1) from (2).

$a^2 - 14a + 2k - a^2 + 11a - k = 0$

$\Rightarrow -3a = -k$

$\Rightarrow a = \frac{k}{3}$

Putting value of a in (1)

$\frac{k^2}{9} - \frac{11}{3}k + k = 0$

$\Rightarrow k^2 - 33k + 9k = 0$

$\Rightarrow k^2 - 24k = 0$

$\Rightarrow k(k - 24) = 0$

As, we have to find non-zero value,

$\therefore k = 24$

56. (B) $\because$ Roots of the equation $x^2 + 3x + 5 = 0$ are non-real.

Thus, given equations will have two common roots

$\Rightarrow \frac{a}{1} = \frac{b}{3} = \frac{c}{5} = \lambda$ [say]

$\therefore$ Thus, minimum value of $a + b + c = 9$ [$\because a, b, c \in N$]

57. (B) Given, $-2x^2 + x + 3$

Since coefficient of x^2 is negative

It will have maximum

$= -\left(\frac{1+24}{4(-2)}\right) = \frac{25}{8}$

Thus range $\left(-\infty, \frac{25}{8}\right]$

58. (B) $x^2 + 2kx + 10 \geq 0 \ \forall x$

This is possible if $D \leq 0$

Thus,

$4k^2 - 4(10) \leq 0$

$k^2 \leq 10$

$-\sqrt{10} \leqslant k \leqslant \sqrt{10}$

$k \in \left[-\sqrt{10}, \sqrt{10}\right]$

59. (C) $(x + a)^2 - (a^2 - 8 - 2a) > 0$

$a^2 - 8 - 2a < 0$

$a^2 - 2a - 8 < 0$

$(a - 4)(a + 2) < 0$

$-2 < a < 4$

60. (B) $\lambda x^2 - 7\lambda x + 8\lambda + 1 > 0 \ \forall \ x \in R$

Then we must have

$\lambda \geq 0$, $D < 0$ ($\lambda = 0$ is valid as it makes LHS = 1 which is greater than 0)

$\Rightarrow \lambda \geq 0$ and $(-7\lambda)^2 - 4\lambda(8\lambda + 1) < 0$

$49\lambda^2 - 32\lambda^2 - 4\lambda < 0$

$17\lambda^2 - 4\lambda < 0$

$\lambda < \frac{4}{17}$

$\Rightarrow 0 \leqslant \lambda < \frac{4}{17}$

61. (B) $Ax^2 + Bx + C = 0$

$1x^2 + 7x + 12 = 0$

$x^2 + 7x + 12 = 0$

$D^2 > 0$

$B^2 > 4ac$

$(7)^2 - 4(1) \times 12 > 0$

Hence, the graph would be concave up intersecting the x-axis at two points.

62. (C) $a < 0$ (as axis is downward parabola)

when roots of equation are opposition in sign

$\frac{a}{c} < 0$

$\Rightarrow c > 0$

63. (D) Since parabola is upward $a > 0$

Curve is crossing x-axis at two points $\Rightarrow$ roots are real $\Rightarrow b^2 - 4ac > 0$

Roots are of opposite signs c/a < 0 ⇒ c < 0

Hence all the options are correct.

64. (B) $3x^2 + 2(k^2 + 1)x + (k^2 - 3k + 2) = 0$

If roots are of opposite sign, then product of roots is negative

Thus,

$\frac{k^2-3k+2}{3} < 0$

$k^2 - 2k - k + 2 < 0$

$(k - 2)(k - 1) < 0$

$1 < k < 2$

65. (C) Given, 2 lies between the roots for $x^2 - ax + a = 0$

Thus,

$f(2) < 0$

$4 - 2a + a < 0$

$4 - a < 0$

$a > 4$

66. (A) Given equation

$x^2 - 2ax + a^2 + a - 3 = 0$

For roots greater than 3

(i) $D > 0$

$4a^2 - 4(a^2 + a - 3) \geq 0$

$3 - a \geq 0$

$a \leq 3$

(ii) $\frac{-b}{2a} > 3$

$\frac{2a}{2} > 3$

$a > 3$

(iii) $f(3) > 0$

$9 - 6a + a^2 + a - 3 > 0$

$a^2 - 5a + 6 > 0$

$(a - 2)(a - 3) > 0$

$a < 2$ or $a > 3$

When we take common

No value of 'a' satisfy

67. (A) Since '2' lies between the roots

Thus,

$f(2) < 0$

$4 + 2(a + 2) - a - 3 < 0$

$4 + 2a + 4 - a - 3 < 0$

$a + 5 < 0$

$a < -5$

$a \in (-\infty , -5)$

68. (C) Given equation $x^2 - (2a - 1)x + a = 0$

If both roots are positive, then

(i) $D \geq 0$

$(2a - 1)^2 - 4a \geq 0$

$4a^2 + 1 - 4a - 4a \geq 0$

$4a^2 - 8a + 1 \geq 0$

$a < \frac{2-\sqrt{3}}{2} or\ a > \frac{2+\sqrt{3}}{2}$

(ii) $-\frac{b}{a} > 0$

$\frac{2a-1}{1} > 0$

$a > \frac{1}{2}$

(iii) $\frac{c}{a} > 0$

$\frac{a}{1} > 0$

$a > 0$

The solution common to all is

$a \in \left(\frac{2+\sqrt{3}}{2}, \infty\right)$

69. (A) $x^2 + (2a + 1)x + (a - 1) = 0$

If one root is greater than 1 and another root is less than 0.

Then,

f(1) < 0 and f(0) < 0

1 + 2a + 1 + a – 1 < 0 and a – 1 < 0

3a + 1 < 0 and a < 1

$a < \frac{-1}{3}$

Thus $a < \frac{-1}{3}$

70. (A) $x^2 - 2ax + a^2 + a - 3 = 0$

For all real and less than 3

(i) D ≥ 0

$4a^2 - 4(a^2 + a - 3) \geq 0$

– a + ≥ 0

a ≤ 3

(ii) $\frac{-b}{2a} < 3$

$\frac{2a}{2} < 3$

a < 3

(iii) f(3) > 0

$9 - 6a + a^2 + a - 3 > 0$

$a^2 - 5a + 6 > 0$

a < 2 or a > 3

Taking common solution

a ∈ (–∞ , 2)

71. (C) for 3 lies outside the roots

f(3) > 0

9 + 6 + k > 0

k > – 15

$2^2 - 4k \geq 0$ and D ≥ 0

$a^2 - 4k \geq 0$

$-4k \geq -4$

$k \leq 1$

$k \in (-15, 1]$

72. (A)

i. $\Delta \geq 0$

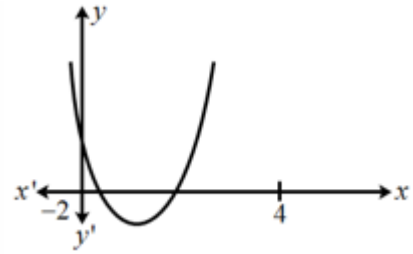
ii. $f(k_1) > 0, f(k_2) > 0$

(iii) $-2 < \frac{-B}{2A} < 4$

i. $(-2m)^2 - 4(m^2 - 1) \geq 1$

$4m^2 - 4m^2 + 4 \geq 0$

$m \in R$

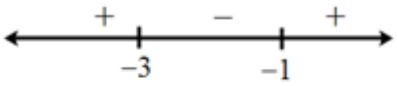


ii. $f(-2) = 4 + 4m + m^2 - 1 > 0$

$m^2 + 4m + 3 > 0$

$(m + 1)(m + 3) > 0$

$m \in (-\infty, -3) \cup (-1, \infty)$



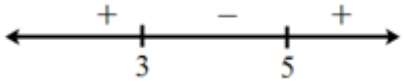


$f(4) = 16 - 8m + m^2 - 1 > 0$

$m^2 - 8m + 15 > 0$

$(m - 3)(m - 5) > 0$

$m \in (-\infty, 3) \cup (5, \infty)$



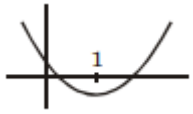
iii. $-2 < \frac{2m}{2} < 4$

$-2 < m < 4$

From Eqns. (i), (ii) and (iii), we get

$m \in (-1, 3)$

73. (C)



$p^2 + p + 1 = a$

is always positive

$\Rightarrow f(1) < 0$

$2p(p + 1) < 0$, so by wavy curve method $p \in (-1, 0)$

74. (B) $\because \alpha + \beta = -b, \alpha\beta = c$ .....(i)

$\because c < 0 \Rightarrow \alpha\beta < 0$

Let $\alpha < 0, \beta > 0$

$\because |\alpha| = -\alpha$ and $\alpha < 0 < \beta$ [$\because \alpha, \beta$] .....(ii)

From eq. (i), we get $-|\alpha| + \beta < 0$

$\Rightarrow \beta < |\alpha|$.....(iii)

From eqs. (ii) and (ii), we get

$\alpha < 0 < \beta < |\alpha|$

75. (D) For roots less than 2

Case I:

Discriminant ≥ 0

$400p^2 - 16\,(25p^2 + 15p - 66) \geq 0$

$\therefore p \leqslant \frac{22}{5}$

Case II:

$-\frac{(-20p)}{2\times 4} < 0$

$\Rightarrow p < \frac{4}{5}$

Case III:

$4 \times f(2) > 0$

$\Rightarrow (16 - 40p + 25p^2 + 15p - 66)\,4 \geq 0$

$\Rightarrow p^2 - p - 2 > 0$

$\Rightarrow p > 2$ or $p < -1$

Combine all cases, we get $p \in (-\infty, -1)$

76. (B) $\because$ Speed of the bus = q km/h

Then the speed of the car = (q + 25) km/h = p km/h

Time taken by the bus to cover 500 km $= \frac{500}{q}h$

and time taken by the car to cover 500 km $= \frac{500}{q+25}h$

According to question, $\frac{500}{q} = \frac{500}{q+25} + 10$

$\Rightarrow 500(q + 25) = 500q + 10q\,(q + 25)$

$\Rightarrow 50(q + 25) = 50q + q(q + 25)$

$\Rightarrow q^2 + 25q - 1250 = 0$

$\Rightarrow (q + 50)(q - 25) = 0$

∴ q = 25, q = –50 (Not possible)

Hence, speed of car = p

= (25 + 25) = 50 km/h

∴ p = 2q

77. (B) Since 2 is the one root of given equation

⇒ 4a + 2b + c = 0 and 2b = a + c (a, b and c are in AP)

⇒ 5a + 2c = 0

⇒ $\frac{c}{a} = \frac{-5}{2}$

Now,

Let α be the other root. Then,

$2 \times \alpha = \frac{c}{a} = -\frac{5}{2}$

$\therefore \alpha = -\frac{5}{4}$

78. (D) $(\alpha^2 + \beta^2)^2 = (\alpha + \beta)(\alpha^3 + \beta^3)$

⇒ $\left(\frac{b^2-2ac}{a^2}\right)^2 = \left(\frac{-b}{a}\right)\left(\frac{-b^3+3abc}{a^3}\right)$

⇒ $4a^2c^2 = acb^2$

⇒ $ac(b^2 - 4ac) = 0$

As $a \neq 0$

⇒ $c\Delta = 0$

79. (B) Let $f(x) = \frac{x^2-2x+3}{x^2-2x-8},\ x \in R - \{-2, 4\}$

Here, the domain of f(x) will be real value of x and for range let,

$y = \frac{x^2-2x+3}{x^2-2x-8}$

⇒ $x^2 - 2x + 3 = y(x^2 - 2x - 8)$

⇒ $x^2(y - 1) - 2x(y - 1) - (8y + 3) = 0$

For x to be real the determinant of equation must be greater than zero

⇒ $b^2 - 4ac > 0$

⇒ $[-2(y - 1)]^2 - 4.(y - 1)(-(8y + 3)) > 0$

⇒ $9y^2 - 7y - 2 > 0$

⇒ (9y + 2)(y – 1) >0

⇒ $y \in \mathbb{R} - \left[\frac{-2}{9}, 1\right]$

80. (B) Let $y = \frac{x^2+2x+4}{x+2}$. Then

$x^2 + x(2 - y) + 2(2 - y) = 0$

As x is real.

$\therefore (2 - y)^2 - 8(2 - y) \geq 0$

⇒ (2 – y)(2 – y – 8) ≥ 0

⇒ (y – 2) (y + 6)≥ 0

⇒ y ≤ –6 or y ≥ 2

Hence, minimum positive value of y is 2.

81. (B) $(2 - \sqrt{3})(2 + \sqrt{3})^{x^2-2x+1} + (2 - \sqrt{3})^{x^2-2x-1}(2 - \sqrt{3}) = 4$

$\Rightarrow (2 + \sqrt{3})^{x^2-2x} + (2 - \sqrt{3})^{x^2-2x} = 4$

$\Rightarrow x^2 - 2x = \pm 1 \Rightarrow x = 1 \mp \sqrt{2}\,.\,1$

82. (C) $\Rightarrow x^2 - 2|x| = \pm 1$

$\Rightarrow |x| = 1 \pm \sqrt{2}.1$, but $|x| \neq 1 - \sqrt{2}$

$\Rightarrow x = \pm(1 + \sqrt{2}).\ \pm 1$

83. (D) $\sqrt{a\sqrt{a\sqrt{a.....}}} = a^{\frac{1}{2}+\frac{1}{4}+\frac{1}{8}.....} = a^{\frac{1/2}{1-1/2}}$

= a = $x^2 - 3$

also $\sqrt{49 + 20\sqrt{6}} = \sqrt{25 + 20\sqrt{6} + 24}$

$= \sqrt{(5 + 2\sqrt{6})^2} = (5 + 6\sqrt{2})$

hence $(5 + 2\sqrt{6})^{x^2-3} + (5 + 2\sqrt{6})^{x^2-x-3-x} = 10$

$x^2 - 3 = \pm 1 \Rightarrow x = \pm 2$ & $x = \pm\sqrt{2}$

domain of $\sqrt{a} \Rightarrow x^2 - 3 > 0 \Rightarrow x > \sqrt{3}$

or $x < -\sqrt{3}$

hence x = 2 only solution

84. (D) Given expression reduce to

$[-(p - q) - (x - p) + (x - q)]^2 = 0$

⇒ $[-p + q - x + p + x - q]^2 = 0$

hence $x \in R$.

85. (C) $y = x^2 + 2a\sqrt{a^2 - 3}x + 4$

$= \left(x + a\sqrt{a^2 - 3}\right)^2 + 4 - a^2(a^2 - 3)$

For perfect square, $4 - a^2 (a^2 - 3) = 0$

$a^4 - 3a^2 - 4 = 0$

$(a^2 - 4) (a^2 + 1) = 0$

⇒ $a = \pm 2$

86. (20) $x^2 + 18x + 45 - 2\sqrt{x^2 + 18 + 45} + 1 = 16$

$\Rightarrow \left(\sqrt{x^2 + 18x + 45} - 1\right)^2 = 16$

$\Rightarrow \sqrt{x^2 + 18 + 45} - 1 = \pm 4$

$\Rightarrow \sqrt{x^2 + 18x + 45} = \pm 4 + 1 = 5, -3$

⇒ $x^2 + 18x + 45 = 25$, (Reject -3)

⇒ $x^2 + 18x + 20 = 0$

Product of root = +20.

87. (D) The inequality $(a + 4)x^2 - 2ax + 2a - 6 < 0$ has to be satisfied for all $x \in R$. Only possible when, coefficient of x^2 and discriminant value are less than zero.

⇒ $a + 4 < 0$ and $4a^2 - 4(a + 4) (2a - 6) < 0$

⇒ $a \in (-\infty , -4)$ and $a^2 + 2a - 24 = (a + 6)(a - 4) > 0$

⇒ $a \in (-\infty , -4)$ and $a \in (-\infty , -6) \cup (4, \infty)$

∴ $a \in (-\infty , -6)$

88. (C) $a\left(\frac{2x+1}{x-1}\right)^2 + b\left(\frac{2x+1}{x-1}\right) + c = 0$

Let $\frac{2x+1}{x-1} - \alpha$

⇒ $x\alpha - \alpha - 2x + 1$

$\Rightarrow x(\alpha - 2) - 1 + \alpha \Rightarrow x = \frac{1+\alpha}{\alpha-2}$

Similarly other roots can be find out.

89. (B) $\frac{1}{2\alpha-3} - x \Rightarrow 2\alpha - 3 - \frac{1}{x} \Rightarrow \alpha - \frac{1}{2}\left(\frac{1}{x} + 3\right)$

$\Rightarrow 2\left(\frac{1+3x}{2x}\right)^2 + 4\left(\frac{1+3x}{2x}\right) - 5 = 0$

$\Rightarrow 11x^2 + 10x + 1 = 0.$

90. (C) $p^3x^2 - pq^2x + q^2r = 0$

91. (D) $x - \frac{1+\alpha}{1-\alpha} \Rightarrow \alpha - \frac{x-1}{x+1}$

$\Rightarrow \left(\frac{x-1}{x+1}\right)^3 - \left(\frac{x-1}{x+1}\right) - 1 = 0$

$\Rightarrow x^3 + 7x^2 - x + 1 = 0 \Rightarrow \prod\left(\frac{1+\alpha}{1-\alpha}\right) = -1.$

92. (B) Let $f(x) = ax^2 + bx + 6$

f(x) = 0, doesn't have two distinct real roots.

So, $D \leq 0$

Also, $f(0) = 6 > 0$

$\Rightarrow f(x) \geq 0 \forall\ x \in R$

$\Rightarrow f(2) \geq 0$

$\Rightarrow 4\alpha + 2b + 6 \geq 0$

$\Rightarrow 2a + b \geq -3$